SKILL DEVELOPMENT (DATA VISUALIZATION USING R)

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6CS52	SKL	0	0	2	1	0	100	100

Course Overview:

Data analysis is the process of data analyzing and extracting useful information which can help in making useful business decisions. As part of this course basics of data analytics are discussed an statistical analysis tools are introduced.

COURSE OBJECTIVES

The course should enable the students to:

1. To learn about the Data Visualization and Process of Visualization.
2. To learn different statistical method for Data visualization.
3. To learn Basics of R and usage of Watson studio
4. To learn Basic Visualization Techniques.
5. Learn about advanced Visualization Techniques.

COURSE OUTCOMES

1. Apply Data Preprocessing for visualization
2. Apply the Inferential statistics and Probability
3. Understand the basics of R Programming and Visualization with R
4. Understand the basic visualization Techniques.
5. Apply the techniques for spatial, Mutlivariate Data

WEEK - 1

1. What is R?, What is R Studio?, RObjects, RDataframes, Write a R Program to plot Student Hall ticket number with student GPD.

WEEK - 2

2. Reading Datasets using R, Import a CSV file in R and check the output.
   Name, Age, Sex
   Shaan, 21, M
   Ritu, 24, F
   Raj, 31, M

WEEK - 3

3. Outliers- a) Consider a Dataframe and remove the missing data in the dataframe.

b) Combining Data sets in R-Write a R program to merge two data frames.

WEEK - 4

4. R Functions- Write a R program to count the number of even numbers in a vector.

WEEK - 5

5. i) The summary of 150 flower data including Sepal Length, Sepal Width, Petal Length and Petal Width. He also wants the summary of Sepal Length vs petal length.

ii) the mean Petal Length of each species.
iii) Write a R Program to segregate the data of flowers having Sepal length greater than 7.
iv) Write a R Programto segregate the data of flowers having Sepal length greater than 7 and Sepal width greater than 3 simultaneously.
v) Write a R Programto view 1st 7 rows of data.
vi) Write a R Program to segregate the data of flowers having Sepal length greater than 7 and Sepal width greater than 3 simultaneously.
vii) Write a R Program to view 1st 7 rows of data.

WEEK - 6

6. Understanding the types of probability.

    i) Find the probability of getting 3 doublets when a pair of fair dice are thrown for 10 times.
    ii) Find the probability of getting 3 or lesser doublets.
    iii) Suppose there are twelve multiple choice questions in an English class quiz. Each question has five possible answers, and only one of them is correct. Find the probability of having four or less correct answers if a student attempts to answer every question at random.

WEEK - 7

7. i) If there are twelve cars crossing a bridge per minute on average, find the probability of having seventeen or more cars crossing the bridge in a particular minute.
    ii) If you throw a dice 20 times then what is the probability that you get following results:
        a. 3 sixes b. b. 6 sixesc. 1,2 and 3 sixes

WEEK - 8

8. i) Write a R Program to calculate permutation for a given values using Functions.
    ii) Write a R Program to calculate Sum of two numbers using function.

WEEK - 9

9. Write a R program to barplot movies, IMBD rating for the given Movie dataset.

WEEK - 10

10. i) Write a R Program to Visualize using histogram for runtime of movies. (Consider Movies dataset)
    ii) Write a R Program to Visualize using Pie chart for genre of movies. (Consider Movies dataset)

TEXT BOOKS

1. R in Action By - Robert L. Kabacoff, Second Edition - Publisher - Dreamtech Press
2. R for Data Science, By - Hadley Wickham and Garrett Gorlemund, O'Reilly

REFERENCE BOOKS

1. R Programming for Data Science by Roger D. Peng (References)
2. The Art of R Programming by Norman Matloff Cengage Learning India.